ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION

CO2 PROBLEM SOLVING ASSIGNMENT

Standards-Compliant Laser Transmitter Design, Wavelength Control and Nonlinearity Management for an 8-Channel DWDM Link

Item	Details
Course Outcome	CO2 – optical sources, modulation dynamics, spectral stability, nonlinear effects, solitons and photonic integration
Assessment	Problem Solving Assignment (100 marks, Bloom's K3–K4)
Student name	R.A. Sanjey Adhithyaa
Register number / Section	192412144
Final recommendation	DFB laser + integrated EAM (EML) in a monolithic InP 8-channel PIC, TEC + wavelength-locker control, 0 dBm per channel, per-span dispersion compensation, no soliton / no DBP

Executive Summary, Assumptions and Contents

This report solves the transmitter-selection and performance-optimisation problem for an 8-channel DWDM link carrying 25 Gb/s NRZ per channel over six amplified 80 km spans of standard single-mode fibre (SMF). Every decision is derived from the calculations that precede it. The key results are collected below and are developed and verified in Stages 1–6.

Quantity	Result	Where
Emission wavelength / internal quantum efficiency	$\lambda = 1.24/0.80 = 1.55$ µm (193.4 THz); $\eta_{int} = 40/(10+40) = 0.80$	Stage 2
Selected source	1550 nm DFB laser with integrated EAM (EML)	Stage 2
Output power at 40 mA	$P = 0.3 \times (40 - 12) = 8.4$ mW = 9.24 dBm	Stage 3
NRZ bandwidth and margin	$B = 0.7 \times 25 = 17.5$ GHz; margin = 20 − 17.5 = 2.5 GHz (14.3 %)	Stage 3
Uncontrolled wavelength drift	0.1 nm/°C × 40 °C = 4.0 nm = 5 channel slots; TEC must hold ±1 °C (design ±0.1 °C)	Stage 3
Effective length and SPM phase	$L_{eff} = 21.17$ km; $\varphi_{NL} = 0.055$ rad/span, 0.330 rad over 6 spans (2 mW)	Stage 4
SBS / SRS thresholds	$P_{th,SBS} = 1.59$ mW (2.0 dBm); $P_{th,SRS} = 1.008$ W (30.0 dBm)	Stage 4
Eye safety	16 mW (12.04 dBm) exceeds the assumed 10 mW limit; design launch 1 mW/ch = 8 mW (9.03 dBm)	Stage 4
Soliton feasibility	$L_D = 11.25$ km; $P_0 = 68.4$ mW; $N = 0.171$ at 2 mW → NOT feasible	Stage 5
Digital backpropagation	Not appropriate for the full 8-channel IM-DD link	Stage 5
Integration	Monolithic InP EML-array PIC on one TEC	Stage 5
Link OSNR at 0 dBm/ch	58 + 0 − 16 − 6 − 7.78 = 28.2 dB (≈ 8 dB above assumed requirement)	Stage 4/6
Final decision	YES – deploy, subject to the listed acceptance tests	Final

Assumptions made where data were not given

- **Chromatic dispersion:** $D = 2\pi c|\beta_2|/\lambda^2 = 15.7$ ps/(nm·km), calculated from the given $|\beta_2| = 20$ ps²/km at 1550 nm (typical SMF value 17 ps/(nm·km) gives the same conclusions).
- **Amplifier and link:** EDFA noise figure (including the mid-stage dispersion-compensating module) NF = 6 dB; span loss 16 dB (0.2 dB/km × 80 km); required OSNR (0.1 nm) for 25 Gb/s NRZ with FEC ≈ 20 dB.
- **Chirp:** linewidth-enhancement factor $\alpha_H = 3$ for direct modulation and $|\alpha_H| \approx 0.5$ for an EAM.
- **Brillouin bandwidth:** $\Delta\nu_B \approx 20$ MHz; Brillouin shift ≈ 11 GHz; extinction ratio of the modulated signal 8 dB.
- **Eye-safety limit:** 10 mW (+10 dBm) is taken as the Class 1 / lowest-hazard-level limit at 1550 nm for an open fibre end (ANSI Z136.2 / IEC 60825-2); the current standard edition must be checked at implementation.
- **Environment:** ambient temperature 0–40 °C (a 40 °C variation); TEC set-point 25 °C. The laser is assumed to emit from one facet and to run at about 1.2 V forward voltage.
- **Fibre:** PMD coefficient 0.1 ps/√km, DCF dispersion −100 ps/(nm·km). Only the SMF is used for nonlinear calculations (the DCF is placed at low power inside the amplifier mid-stage).

Contents

Section	Marks	Topic
Stage 1	10	Interpret the communication requirement; Problem Requirement Statement
Stage 2	15	Investigate and select the optical source (Design Decision 1)
Stage 3	20	Modulation, chirp and spectral stability (Design Decision 2)
Stage 4	20	Optical nonlinearities and eye safety (Design Decision 3)
Stage 5	20	Soliton, digital backpropagation, standards and integration (Design Decision 4)
Stage 6	5	Verification table and critical evaluation
Final design	10	Integrated transmitter, block diagram, parameter table
Decision	–	YES / NO with technical justification

Stage 1 – Interpret the Communication Requirement

The design is worked backward from what the link must deliver: 8 × 25 Gb/s = 200 Gb/s of aggregate capacity over 6 × 80 km = 480 km, with every channel regenerated only optically. The transmitter is therefore the first link in a chain whose later stages (fibre loss, amplifier noise, dispersion, nonlinearity) can only degrade what it launches.

1.1 The DWDM channel plan

With 100 GHz spacing the channels sit on the ITU-T G.694.1 frequency grid. Eight consecutive channels starting at 193.1 THz are chosen so that the band is centred close to 1550 nm. Wavelengths follow $\lambda = c/f$ with $c = 2.998 \times 10^8$ m/s.

Channel	Frequency (THz)	Vacuum wavelength (nm)	Offset from 1550 nm (nm)
λ1	193.1	1552.52	+2.52
λ2	193.2	1551.72	+1.72
λ3	193.3	1550.92	+0.92
λ4	193.4	1550.12	+0.12
λ5	193.5	1549.32	−0.68
λ6	193.6	1548.51	−1.49
λ7	193.7	1547.72	−2.28
λ8	193.8	1546.92	−3.08

Spacing of 100 GHz corresponds to $\Delta\lambda = \lambda^2\Delta f/c = (1550\text{ nm})^2 \times 100\text{ GHz} / 2.998 \times 10^8\text{ m/s} \approx 0.80$ nm, and the whole 8-channel band spans 7 × 0.8 = 5.6 nm, well inside the gain window of an erbium-doped fibre amplifier (C-band, about 1530–1565 nm).

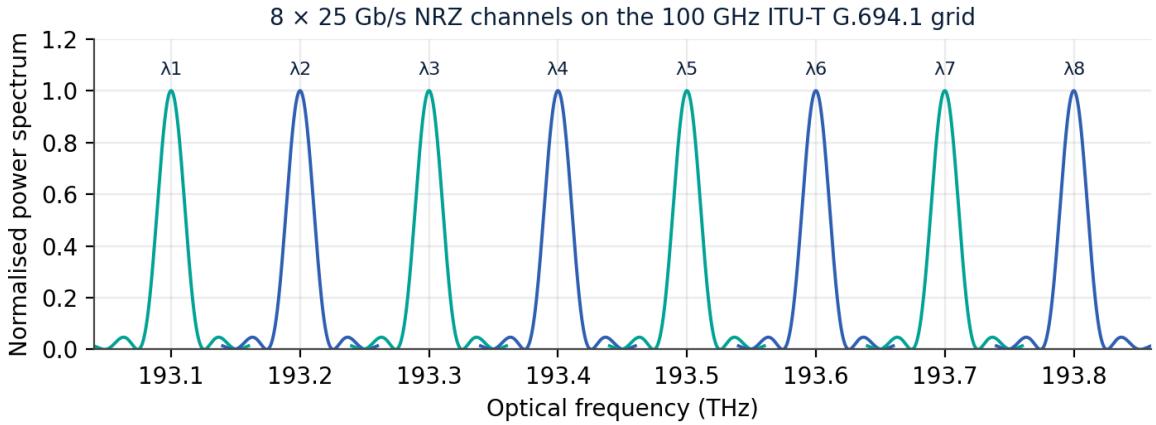


Figure 1 – Eight 25 Gb/s NRZ spectra on the 100 GHz grid; main lobes (±25 GHz) fit inside each slot with little overlap.

1.2 Why speed, narrow linewidth and power are all required

High modulation speed. A 25 Gb/s NRZ stream needs an analogue bandwidth of $B \approx 0.7\ R_b = 17.5$ GHz. An LED, whose bandwidth is limited by the carrier lifetime to a few hundred MHz, cannot do this; a laser (or a laser plus external modulator) must supply at least 17.5 GHz with margin for the driver and package.

Narrow linewidth and spectral purity. The signal spectrum is about 25 GHz (0.2 nm) wide even for an ideal source. Any additional source width multiplies through chromatic dispersion: the pulse broadening is $\Delta T = D \cdot L \cdot \Delta\lambda$. For an 80 km span with D = 15.7 ps/(nm·km):

$$\Delta T_{FP} = 15.7 \times 80 \times 3\ \text{nm} \approx 3.8\ \text{ns} \quad (\text{FP laser, 3 nm multimode}) \gg T_b = 40\ \text{ps}$$

$$\Delta T_{DFB,min} = 15.7 \times 80 \times 0.2\ \text{nm} \approx 251\ \text{ps} \quad (\text{data-limited width only})$$

So a single-longitudinal-mode source is indispensable, and even then the dispersion accumulated per span (15.7 × 80 = 1256 ps/nm) must be compensated. In addition, side modes must be suppressed (SMSR ≥ 30 dB) so that they do not leak into neighbouring 0.8 nm slots.

High (but not excessive) output power. Each span loses 16 dB, so the signal-to-noise ratio is set by the launch power into the first span and by the EDFA noise. The usable range is bounded above by SBS (Stage 4), by eye safety and by Kerr nonlinearity. The transmitter must therefore deliver enough power for good OSNR yet allow it to be set accurately and conservatively.

1.3 Source characteristics that set wavelength, efficiency and purity

Characteristic	What it determines	Relevance to this link
Bandgap E_g	Emission wavelength $\lambda = 1.24/E_g(\text{eV})$ µm	E_g = 0.80 eV gives 1.55 µm, inside the EDFA band
Internal quantum efficiency η_{int}	Fraction of carriers that recombine radiatively = $\tau_{nr}/(\tau_r+\tau_{nr})$	Sets how much injected current becomes photons
Threshold current I_{th}	Current at which gain equals loss	12 mA; bias well above it removes turn-on delay
Slope efficiency η_s	dP/dI above threshold	0.3 mW/mA gives 8.4 mW at 40 mA
Linewidth Δν, SMSR	Spectral purity, phase noise, SBS threshold, crosstalk	MHz-class linewidth; SMSR ≥ 30–35 dB
Modulation bandwidth	Highest data rate before ISI	20 GHz given vs 17.5 GHz needed

1.4 Modulation, chirp and temperature quantities to be controlled

- **Relaxation-oscillation frequency f_R** and damping, which set the usable direct-modulation bandwidth and the large-signal ringing.
- **Chirp and α_H:** frequency excursions $\Delta\nu(t) = (\alpha_H/4\pi)[d \ln P/dt + \kappa P(t)]$ that broaden pulses through dispersion.
- **Extinction ratio and bias:** a bias too close to threshold creates turn-on delay and pattern-dependent jitter; too high reduces ER.
- **Wavelength temperature coefficient** (0.1 nm/°C, i.e. ≈ 12.5 GHz/°C), which for 40 °C of ambient swing exceeds the 0.8 nm channel spacing by a factor of five.
- **Relative intensity noise (RIN)** and optical feedback, which affect the Q-factor of 25 Gb/s NRZ.

1.5 Nonlinear quantities that matter for multichannel, multispan operation

Because the link has many channels, many spans and a non-trivial launch power, the quantities that become important are the effective length L_{eff}, the nonlinear coefficient γ, the nonlinear phase $\varphi_{NL} = \gamma P L_{eff}$ (accumulating over spans), the channel walk-off ($D \cdot \Delta\lambda$), the phase mismatch Δβ that decides four-wave mixing, and the Brillouin and Raman thresholds. With N = 8 equally spaced channels, four-wave mixing generates $(N^3 - N^2)/2 = (512 - 64)/2 = 224$ mixing products, many falling exactly on other channels. These are evaluated numerically in Stage 4.

1.6 Role of the standards

Standard	Role in defining the transmitter specification
ITU-T G.957 (with G.959.1, G.694.1, G.698.x)	Defines optical-interface parameters for SDH-class links: mean launched power range, central wavelength range, spectral width (RMS / −20 dB), minimum SMSR (30 dB), minimum extinction ratio, eye mask. G.694.1 gives the 100 GHz grid and G.698.x/G.959.1 the multichannel DWDM interface rules.
IEEE 802.3bm / 802.3bs	Define 25 Gb/s-per-lane PHYs in 100G, 200G and 400G Ethernet: lane wavelength window, average launch power, OMA, extinction ratio, SMSR, RIN and transmitter-and-dispersion-eye-closure (TDECQ) limits, plus RS-FEC.
ANSI Z136 (Z136.2 for fibre communication)	Sets maximum permissible exposure and hazard classes; fixes the total optical power that may leave an open connector and requires labelling, automatic laser shutdown (ALS) and controlled access above the lowest class.

1.7 System-level parameters influencing soliton / DSP compensation and integration

Soliton feasibility depends on T_0, $|\beta_2|$, γ, launch power and the amplifier spacing relative to L_D. DSP-based compensation depends on whether the receiver is coherent, on the number of channels that must be processed jointly, and on the nonlinear phase actually present. Integration depends on channel count, thermal tracking between lasers and the multiplexer, coupling losses, reliability and volume.

1.8 Characteristics of an acceptable transmitter

- Single-longitudinal-mode emission on the ITU grid at 1546.9–1552.5 nm with SMSR ≥ 35 dB and a MHz-class linewidth.
- Modulation bandwidth > 17.5 GHz with practical margin and low chirp.
- Wavelength held to ±0.1 nm over a 40 °C ambient range.
- Per-channel launch low enough for SBS, SPM, XPM and FWM to be negligible, and total launch within the eye-safety limit.
- Compliance with the G.957 / G.694.1 / IEEE 802.3bm-bs parameter set; automatic laser shutdown.
- Compact, reliable, scalable implementation.

> **Problem Requirement Statement**
>
> The proposed 8-channel DWDM transmitter shall launch eight single-mode, ITU-grid-locked (193.1–193.8 THz, ±0.1 nm) 25 Gb/s NRZ signals with SMSR ≥ 35 dB, extinction ratio ≥ 8 dB, modulation bandwidth > 17.5 GHz and low chirp into a 6 × 80 km SMF link at a per-channel power that keeps SBS below threshold (< 1.59 mW), the SPM phase ≪ 1 rad, and the total launched power within the 10 mW eye-safety limit, while retaining ≥ 20 dB OSNR at the receiver and meeting ITU-T G.957/G.694.1 and IEEE 802.3bm/bs transmitter requirements in a compact, thermally stable implementation.

Stage 2 – Investigate and Select the Optical Source

2.1 Recombination processes and the role of the bandgap

In the active region electrons and holes injected across the p–n heterojunction recombine by several competing routes. **Radiative recombination** (spontaneous and stimulated) emits a photon of energy hν ≈ E_g; **non-radiative recombination** (Shockley–Read–Hall defect trapping, Auger and surface recombination) converts the energy to heat. The carrier-density rate equation below threshold is

$$dN/dt = I/(qV) - N/\tau_r - N/\tau_{nr}$$

The bandgap fixes the photon energy and therefore the wavelength; for lattice-matched InGaAsP/InP an E_g near 0.80 eV puts the emission in the 1.55 µm low-loss window of silica fibre and in the gain band of the EDFA.

Calculation 2.1 – Emission wavelength

$$\lambda = 1.24 / E_g(\text{eV}) = 1.24 / 0.80 = 1.55\ \mu\text{m} = 1550\ \text{nm}$$

$$f = c/\lambda = 2.998 \times 10^{8} / 1.55 \times 10^{-6} = 1.934 \times 10^{14}\ \text{Hz} = 193.4\ \text{THz}$$

The emission frequency of 193.4 THz lies exactly on an ITU grid point (channel λ4). The 100 GHz channel spacing corresponds to an energy step of $h\Delta f = 6.626 \times 10^{-34} \times 10^{11} / 1.602 \times 10^{-19} = 0.41$ meV, so the other channels are reached by tiny changes of the Bragg period rather than by new materials.

Calculation 2.2 – Internal quantum efficiency

$$1/\tau = 1/\tau_r + 1/\tau_{nr} = 1/10\ \text{ns} + 1/40\ \text{ns} = 0.125\ \text{ns}^{-1} \rightarrow \tau = 8\ \text{ns}$$

$$\eta_{int} = (1/\tau_r) / (1/\tau_r + 1/\tau_{nr}) = \tau/\tau_r = 8/10 = 0.80\ \ (80\ \%)$$

Calculation 2.3 – External efficiency and wall-plug efficiency

$$\eta_d = \eta_s \cdot (q/h\nu) = 0.3\ \text{W/A} \div 0.80\ \text{V} = 0.375\ \ (37.5\ \%)$$

The differential efficiency is smaller than $\eta_{int} = 0.80$ because only the fraction $\alpha_m/(\alpha_m+\alpha_{int}) = 0.375/0.80 = 0.47$ of the generated photons leaves through the output facet (the rest is absorbed or lost to the back facet). With V ≈ 1.2 V (assumed) the electrical input is 40 mA × 1.2 V = 48 mW and the wall-plug efficiency is 8.4/48 = 17.5 %.

2.2 Threshold condition, linewidth and side-mode suppression

Laser oscillation begins when the round-trip modal gain equals the total loss:

$$\Gamma g_{th} = \alpha_{int} + (1/2L) \cdot \ln(1/R_1R_2) \quad \text{(Fabry–Perot)}$$

In a DFB laser the distributed grating replaces the mirror term by a coupling coefficient κ, and only the mode at the Bragg wavelength $\lambda_B = 2n_{eff}\Lambda/m$ has the lowest threshold. With $n_{eff} \approx 3.2$ and first order (m = 1), Λ = 1550/(2 × 3.2) ≈ 242 nm. The measured threshold I_{th} = 12 mA corresponds to a threshold carrier density N_{th} through $I_{th} = qVN_{th}/\tau$.

The intrinsic (Schawlow–Townes–Henry) linewidth is $\Delta\nu = v_g^2 h\nu\ g\ n_{sp}\alpha_m(1 + \alpha_H^2) / (8\pi P_0)$; it falls as 1/P and is a few MHz for a DFB at several mW. For an intensity-modulated 25 Gb/s link, MHz linewidth is negligible compared with the 25 GHz signal bandwidth, but it controls three effects that matter here: (i) the Brillouin threshold, which is lowered when the carrier is narrower than $\Delta\nu_B \approx 20$ MHz; (ii) phase-noise-to-intensity-noise conversion through dispersion; (iii) any later coherent upgrade.

The **side-mode suppression ratio** (SMSR) is the power ratio of the main mode to the strongest side mode. In an 8-channel link a side mode of one laser falls into the passband of the neighbouring channel and appears as in-band crosstalk. SMSR ≥ 30 dB (both ITU-T and IEEE) keeps the crosstalk ≤ −30 dB;

a DFB typically achieves 35–50 dB, whereas an FP laser has SMSR < 10 dB and suffers mode-partition noise.

2.3 Comparison of candidate sources

Parameter	LED	FP laser	DBR laser	DFB laser	VCSEL (1550)	QD laser
Fibre-coupled power	−20 to −10 dBm	0–8 dBm	0–10 dBm	3–10 dBm	−3 to +3 dBm	0–8 dBm
Spectral width	50–100 nm	1–5 nm, multimode	MHz line; SMSR > 40 dB	MHz line; SMSR 35–50 dB	10–50 MHz	MHz-class (low α_H)
Direct modulation	< 0.5 GHz	10–20 GHz	5–15 GHz	10–25 GHz (EML 30–50)	5–15 GHz	10–25 GHz (research)
Wall-plug efficiency	1–5 %	10–25 %	10–20 %	15–30 %	10–15 %	15–30 %
Threshold behaviour	None	10–30 mA	15–40 mA	5–20 mA	1–5 mA	Very low; T-insensitive
Fibre coupling to SMF	Poor (< 2 %)	Fair	Good	Good	Excellent (round beam)	Good
DWDM suitability	Unsuitable	Poor (mode partition)	Good, tunable	Excellent	Limited by power at 1550 nm	Promising, immature

The LED and the FP laser fail on spectral width, mode-partition noise and (for the LED) speed. The VCSEL's output power at 1550 nm and its tuning mechanism are not yet adequate for an 80 km multi-span link at 25 Gb/s per channel. A DBR laser is a possible choice but needs multi-section control and is prone to mode hops; the QD laser is attractive for temperature insensitivity and low chirp but is not yet mature for 1550 nm commercial deployment. The DFB satisfies all constraints simultaneously.

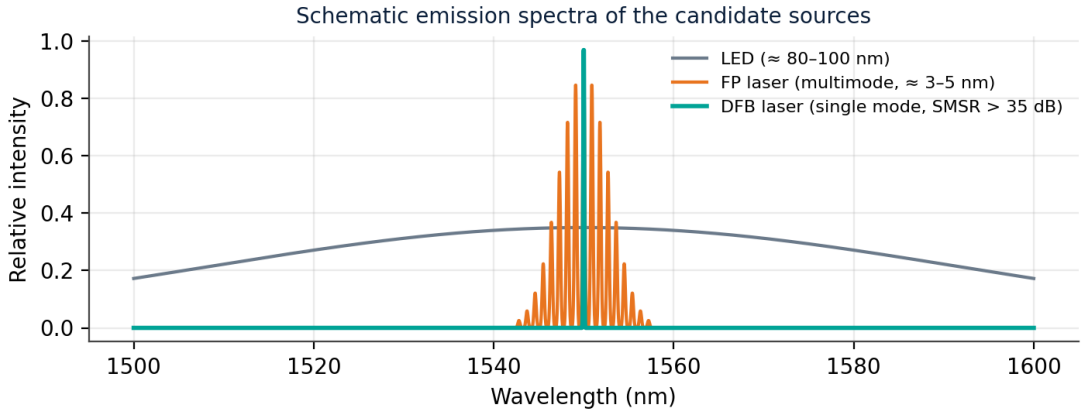


Figure 2 – Schematic emission spectra of LED, FP and DFB sources around 1550 nm.

2.4 Selection and justification

Technical parameter	DFB result	Requirement	Verdict
Wavelength / frequency	1.55 µm, 193.4 THz (on grid)	1546.9–1552.5 nm	Met
Internal quantum efficiency	0.80; η_d = 37.5 %	High efficiency	Met
Output power at 40 mA	8.4 mW (9.24 dBm)	Enough for EAM + mux + EDFA budget	Met, 10 dB spare
Modulation bandwidth	20 GHz	> 17.5 GHz	Met (+2.5 GHz); EAM adds margin
Spectral purity	Single mode, SMSR > 35 dB, MHz linewidth	SMSR ≥ 30 dB	Met
Temperature behaviour	0.1 nm/°C	±0.1 nm over 40 °C	Needs TEC (Stage 3)

Technically credible alternative

The **QD laser** is the best alternative. Its very low and temperature-insensitive threshold, small linewidth-enhancement factor ($\alpha_H \approx 0$–1) and low chirp would allow uncooled or lightly cooled directly-modulated operation, eliminating the TEC and its roughly 1–2 W of power. It becomes preferable when (a) 1550 nm QD DFB devices reach commercial yield and 25 Gb/s direct-modulation bandwidth, and (b) the system is power-limited (for example a dense data-centre interconnect) rather than reach-limited. A tunable DBR laser would be a second alternative if one spare part must serve all eight wavelengths.

> **DESIGN DECISION 1 – Optical source**
>
> I recommend **a 1550 nm DFB laser with an integrated electro-absorption modulator (EML)** optical source because its emission at 1.55 µm (193.4 THz) with $\eta_{int} = 0.80$ sits on the ITU grid and in the EDFA band, it delivers 8.4 mW (9.24 dBm) at 40 mA with 37.5 % differential efficiency, its single-mode spectrum (SMSR > 35 dB, MHz linewidth) fits inside the 0.8 nm slot, and its 20 GHz bandwidth exceeds the 17.5 GHz NRZ requirement, with the EAM providing further bandwidth and low chirp for 80 km spans.

Stage 3 – Modulation, Chirp and Spectral-Stability Strategy

3.1 Output power at the stated bias

$$P = \eta_s(I - I_{th}) = 0.3\ \text{mW/mA} \times (40\ \text{mA} - 12\ \text{mA}) = 0.3 \times 28 = 8.4\ \text{mW}$$

$$P(\text{dBm}) = 10 \log_{10}(8.4\ \text{mW} / 1\ \text{mW}) = 10 \times 0.9243 = 9.24\ \text{dBm}$$

The bias is 40/12 = 3.33 times threshold, comfortably above the region of turn-on delay. The L–I characteristic around the operating point is:

Bias I (mA)	$I - I_{th}$ (mA)	P (mW)	P (dBm)
20	8	2.4	3.80
30	18	5.4	7.32
40 (design)	28	8.4	9.24
50	38	11.4	10.57

Power budget through the transmitter (assumed losses in brackets): DFB chip 9.24 dBm → EAM average insertion loss (−6 dB) → 3.24 dBm → chip-to-waveguide coupling (−1.5 dB) → 1.74 dBm → 8:1 AWG multiplexer (−4 dB) → −2.26 dBm → booster EDFA gain (+2.26 dB) → **0 dBm (1 mW) per channel**, i.e. 8 mW = 9.03 dBm in total. The 9.24 dBm available from the laser therefore leaves ample headroom.

3.2 Bandwidth requirement and margin

$$B = 0.7\ R_b = 0.7 \times 25\ \text{Gb/s} = 17.5\ \text{GHz}$$

$$\text{Margin} = 20\ \text{GHz} - 17.5\ \text{GHz} = 2.5\ \text{GHz}\ \ (20/17.5 = 1.143, \text{i.e. } 14.3\ \%\ \text{or } 0.58\ \text{dB})$$

The candidate bandwidth is sufficient but the margin is small. If the Ethernet line rate of 25.78125 Gb/s (64B/66B coding) is used, B = 18.05 GHz and the margin falls to 1.95 GHz (10.8 %). Package parasitics, bond-wire inductance and driver roll-off easily remove 2 GHz, so a directly modulated laser would be marginal; this is a strong reason to consider external modulation.

3.3 Rate equations and relaxation oscillation

The single-mode carrier and photon rate equations are

$$dN/dt = I/(qV) - N/\tau_e - v_g\, a\, (N - N_0)\, S / (1 + \varepsilon S)$$

$$dS/dt = \Gamma\, v_g\, a\, (N - N_0)\, S / (1 + \varepsilon S) - S/\tau_p + \Gamma\, \beta\, N/\tau_r$$

where N is the carrier density, S the photon density, τ_p the photon lifetime, a the differential gain, ε the gain-compression factor and β the spontaneous-emission factor. In steady state above threshold $S_0 \approx \tau_p(I - I_{th})/(qV)$.

Linearising for small modulation gives a damped second-order response with relaxation-oscillation (resonance) frequency

$$f_R = (1/2\pi)\cdot\sqrt{(\Gamma\, v_g\, a\, S_0 / \tau_p)} \propto \sqrt{(I - I_{th})}$$

$$H(f) = f_R^2 / (f_R^2 - f^2 + j\, f\, \gamma/2\pi), \quad f_{3dB} \approx 1.55\ f_R \text{ (low damping)}$$

Using the given f_{3dB} = 20 GHz and a damping rate $\gamma/2\pi \approx 9$ GHz, the model in Figure 3 requires $f_R \approx$ 13.9 GHz. Because $f_R \propto \sqrt{(I - I_{th})}$, the bandwidth at other bias currents scales as

$$f_{3dB}(I) = 20\ \text{GHz} \times \sqrt{[(I - 12\ \text{mA})/28\ \text{mA}]}\ \geq 17.5\ \text{GHz}\ \rightarrow\ I \geq 12 + 28(0.875)^2 = 33.4\ \text{mA}$$

Thus a directly modulated version of this laser cannot be biased below about 33 mA without violating the 17.5 GHz requirement; 40 mA gives the stated 14 % margin.

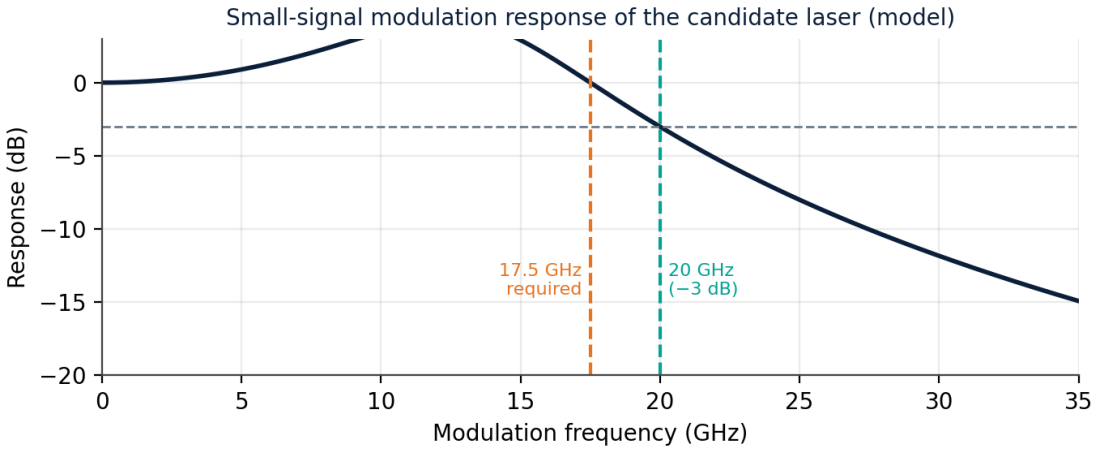


Figure 3 – Modelled small-signal response (f_R = 13.9 GHz, $\gamma/2\pi$ = 9 GHz) with the 17.5 GHz requirement.

Small-signal versus large-signal modulation

Aspect	Small-signal	Large-signal (25 Gb/s NRZ)
Drive	Modulation depth $\lesssim$ 10–20 % of $(I - I_{th})$; equations linearised	Full 0/1 swing; rate equations are nonlinear
Response	Smooth second-order H(f) with resonance peak at f_R	Turn-on delay $t_d = \tau \ln[I_p/(I_p+I_b-I_{th})]$, ringing at f_R, bit-pattern-dependent jitter
Bandwidth	$f_{3dB} \approx 1.55\ f_R$	Effective bandwidth is lower; gain compression ε damps the ringing but limits the speed
Chirp	Sinusoidal, small	Large transient and adiabatic chirp; spectrum broadens
Use	Specifying and testing the device	Eye-diagram design: bias, swing and ER

3.4 Direct versus external modulation

The instantaneous frequency deviation of a directly modulated laser is

$$\Delta\nu(t) = (\alpha_H/4\pi)\ [\ (1/P)(dP/dt) + \kappa\, P(t)\ ]$$

The first term is the transient chirp and the second the adiabatic chirp. Taking $\alpha_H = 3$ and a rise time $t_r \approx 0.35/20$ GHz = 17.5 ps with a full power swing:

$$\Delta\nu_{trans} \approx (3/4\pi) \times (1/17.5\ \text{ps}) = 1.36 \times 10^{10}\ \text{Hz} = 13.6\ \text{GHz}$$

$$\Delta\lambda = \lambda^2\Delta\nu/c = (1.55 \times 10^{-6})^2 \times 13.6 \times 10^9 / 2.998 \times 10^8 = 0.109\ \text{nm}$$

$$\Delta T_{chirp} = D\ L\ \Delta\lambda = 15.7 \times 80 \times 0.109 = 137\ \text{ps} \approx 3.4\ \text{bit periods}$$

A directly modulated laser would therefore close the eye completely over a single uncompensated 80 km span; its spectrum (25 GHz data + 2 × 13.6 GHz chirp ≈ 50 GHz) would also occupy half of the 100 GHz slot. Even for ideal, chirp-free NRZ the dispersion-limited length is

$$L_{max} \approx 104\,000 / (D\ B^2) = 104\,000 / (15.7 \times 25^2) \approx 10.6\ \text{km}$$

so 80 km spans require dispersion compensation (about −1256 ps/nm per span, i.e. 12.6 km of −100 ps/(nm·km) DCF or a chirped fibre Bragg grating) irrespective of the source. With compensation, a low-chirp external modulator is still decisive because residual dispersion, SPM and the sign of the chirp interact over six spans.

Property	Direct modulation (DML)	EAM (integrated EML)	MZ modulator
Chirp (α_H)	+2 to +5, 0.1 nm transient	−1 to +0.5, ≈ 0.02 nm	≈ 0 (balanced push-pull)
Bandwidth	20 GHz; 14 % margin only	30–50 GHz	30–40 GHz

Property	Direct modulation (DML)	EAM (integrated EML)	MZ modulator
Extinction ratio	4–6 dB	8–10 dB	> 12 dB
Insertion loss / cost	None / lowest	≈ 6 dB, monolithic with DFB	4–6 dB, larger, higher drive voltage
Power and size	Smallest, lowest drive power	Small, ≈ 2 Vpp	Largest, 3–6 V
Reach at 25 Gb/s	≲ 2–3 km uncompensated	≈ 10 km uncompensated; 80 km with DCM	≥ 80 km with DCM

The EAM is chosen: it integrates monolithically with the DFB section on InP (no extra coupling), supplies > 30 GHz bandwidth, and its chirp is small and can be negative, which extends the reach on anomalous-dispersion fibre. A Mach–Zehnder modulator would give even lower chirp but at the cost of footprint, drive power and insertion loss that are unnecessary for NRZ at 25 Gb/s.

3.5 Wavelength drift, TEC stability and tuning methods

$$\Delta\lambda = (d\lambda/dT) \times \Delta T = 0.1\ \text{nm/°C} \times 40\ \text{°C} = 4.0\ \text{nm}$$

$$\Delta\lambda\ /\ \text{channel spacing} = 4.0\ \text{nm} / 0.8\ \text{nm} = 5.0\ \text{channel slots} \quad (\approx 499\ \text{GHz} = c\Delta\lambda/\lambda^2)$$

An uncontrolled laser would therefore walk across five neighbouring channels and collide with them. The required temperature stability for the tolerance ±0.1 nm is

$$\Delta T_{\text{allowed}} = \pm 0.1\ \text{nm} / (0.1\ \text{nm/°C}) = \pm 1.0\ \text{°C} \quad (2\ \text{°C peak-to-peak})$$

That consumes the entire error budget, so a design target of ±0.1 °C (±0.01 nm) is adopted, leaving room for the locker, ageing and thermal gradients:

Contribution to wavelength error	Budget (nm)
TEC residual (±0.1 °C × 0.1 nm/°C)	±0.010
Wavelength-locker accuracy	±0.020
Ageing over life (end-of-life drift)	±0.030
Thermal gradient / self-heating	±0.010
Total (worst case, arithmetic)	**±0.070 nm < ±0.100 nm**

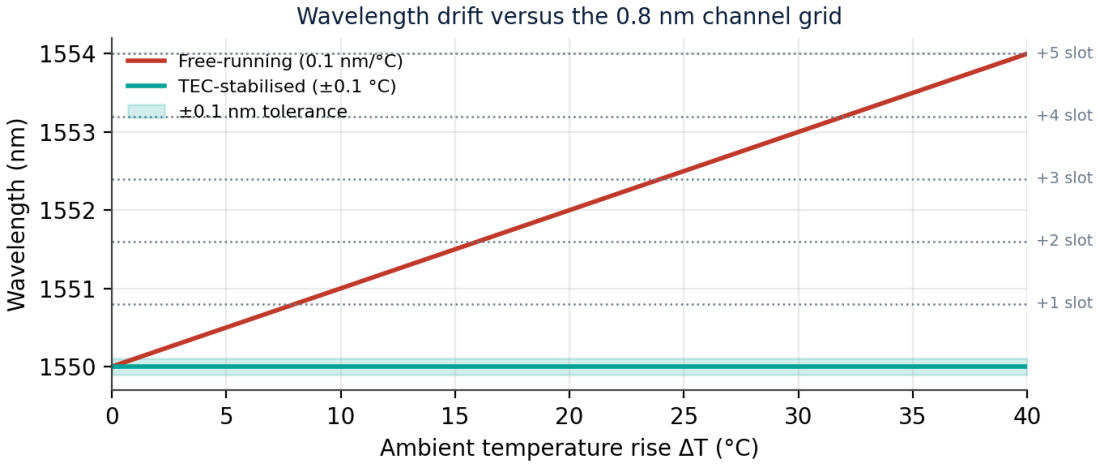


Figure 4 – Uncontrolled drift crosses five channel slots over 40 °C; TEC control keeps the wavelength within ±0.1 nm.

With the TEC set at 25 °C and ambient 0–40 °C, the TEC cools by up to 15 °C or heats by up to 25 °C. The per-channel dissipation is the electrical power minus optical power, 48 − 8.4 ≈ 40 mW plus the EAM; for the array a TEC of ≈ 2 W is adequate.

Criterion	TEC (temperature) tuning	Injection-current tuning
Coefficient	≈ 0.1 nm/°C (12.5 GHz/°C)	≈ 0.005 nm/mA (assumed, thermal)
Range	Several nm	< 0.3 nm
Speed	Seconds (thermal time constant)	µs–ms
Side effects on signal	None on power, chirp, ER	P changes by 0.3 mW/mA; ER, chirp and bandwidth change
To correct 0.1 nm	1 °C	20 mA → ΔP = 6 mW (unacceptable)
Conclusion	Coarse + fine control, closed loop with locker	Only micro-trim; not used

DESIGN DECISION 2 – Modulation and wavelength control

I recommend **external (integrated EAM) modulation** and **TEC-based wavelength control with a closed-loop wavelength locker (PID, ±0.1 °C)** because the laser gives 8.4 mW (9.24 dBm), the 20 GHz direct bandwidth leaves only a 2.5 GHz (14.3 %) margin over the 17.5 GHz NRZ requirement, direct-modulation chirp (≈ 13.6 GHz, 0.109 nm) would broaden pulses by 137 ps over one span (3.4 bit periods), and an uncontrolled 4.0 nm drift (5 channel slots) must be reduced to ≤ ±0.01 nm without altering optical power or chirp, which only TEC temperature tuning achieves.

Stage 4 – Investigate and Control the Optical Nonlinearities

4.1 Effective length and SPM phase shift

The fibre loss of 0.2 dB/km is converted to a field-power attenuation coefficient $\alpha = 0.2/4.343 = 0.04605$ km^{-1}.

$$\alpha L = 0.04605 \times 80 = 3.684; \quad e^{-\alpha L} = 0.02512$$

$$L_{eff} = (1 - e^{-\alpha L})/\alpha = (1 - 0.02512)/0.04605 = 21.17 \text{ km}$$

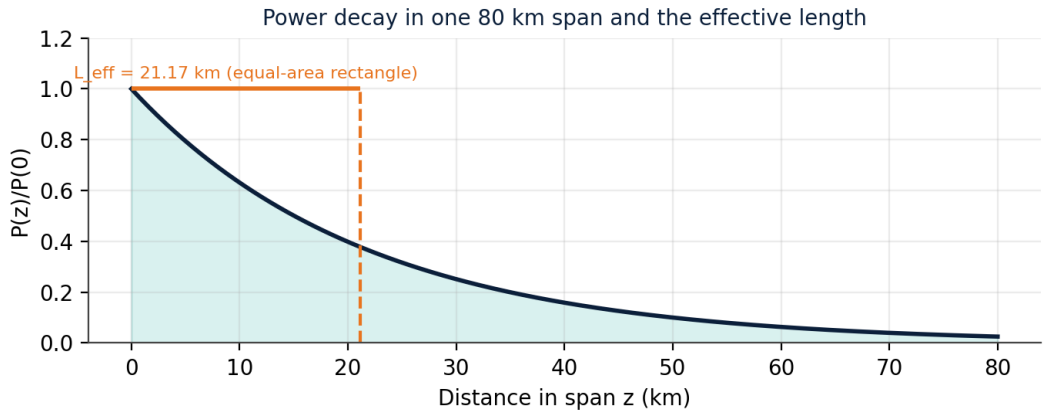


Figure 5 – Signal power along one span; the shaded area equals the rectangle of height P(0) and length L_{eff}.

The nonlinear phase generated by self-phase modulation in one span is $\varphi_{NL} = \gamma P L_{eff}$. For P = 2 mW:

$$\varphi_{NL,span} = 1.3 \text{ W}^{-1}\text{km}^{-1} \times 0.002 \text{ W} \times 21.17 \text{ km} = 0.0550 \text{ rad}$$

$$\varphi_{NL,6\ spans} = 6 \times 0.0550 = 0.330 \text{ rad} \quad \text{(upper bound, coherent accumulation)}$$

This is well below the ≈ 1 rad commonly taken as the onset of significant SPM penalty. At the design launch of 1 mW the values halve to 0.0275 rad per span and 0.165 rad for six spans. SPM alone is therefore not the limiting impairment.

4.2 XPM and FWM in the 100 GHz grid

Cross-phase modulation (XPM) arises because the intensity of one channel changes the refractive index seen by its neighbours. In the worst case (all 7 other channels at 2 mW, aligned) the phase shift is twice the SPM value from each, so

$$\varphi_{XPM,max} = 2\gamma(7P)L_{eff} = 2 \times 1.3 \times 0.014 \times 21.17 = 0.771 \text{ rad per span}$$

However XPM acts only while the channels overlap in time. The walk-off per unit length between adjacent channels is $d_{12} = D \cdot \Delta\lambda = 15.7 \times 0.8 = 12.6$ ps/km, so the walk-off length for a 40 ps bit is $L_w = 40/12.6 = 3.2$ km, much less than L_{eff}. The XPM efficiency at the 12.5 GHz half-bit-rate frequency is

$$|H_{XPM}| \approx \alpha/\sqrt{(\alpha^2 + (2\pi f d_{12})^2)} = 4.6\times10^{-5} / \sqrt{[(4.6\times10^{-5})^2 + (9.9\times10^{-4})^2]} \approx 0.047$$

so the effective XPM phase falls to about 0.77 × 0.047 = 0.036 rad per span, a −26 dB reduction. Dispersion thus converts the potentially severe XPM into a small intensity-noise penalty; it does, however, convert phase noise to amplitude distortion, so the DCF must not be so strong that the residual dispersion goes to zero locally.

Four-wave mixing (FWM) generates new frequencies $f_{ijk} = f_i + f_j - f_k$. With equal spacing, products overlap on the signal channels and appear as crosstalk. The efficiency depends on the phase mismatch

$$\Delta\beta = 2\pi\lambda^2 D(\Delta f)^2/c = 2\pi \times (1.55\times10^{-6})^2 \times 1.568\times10^{-5} \times (10^{11})^2 / 2.998\times10^{8} = 7.9 \text{ rad/km}$$

$$\eta_{FWM} \approx \alpha^2/(\alpha^2 + \Delta\beta^2) = 0.0461^2 / (0.0461^2 + 7.9^2) = 3.4 \times 10^{-5} \ (-44.7 \text{ dB})$$

$$P_{FWM} = \eta\,(d\,\gamma\,L_{eff})^2\,P^3 = 3.4\times10^{-5} \times (6 \times 1.3 \times 21.17)^2 \times (2\times10^{-3})^3 = 7.5 \times 10^{-9}\ \text{W} \approx -51\ \text{dBm}$$

(with degeneracy factor d = 6). Relative to a 3 dBm carrier, this is about −54 dB per term; allowing six coherent spans (+15.6 dB) and three overlapping products (+5 dB) gives a worst-case crosstalk near −33 dB at 2 mW, and −39 dB at 1 mW because the crosstalk relative to the carrier scales as P^2 (−6 dB). Both are acceptable, which is possible only because the high local dispersion of SMF (D = 15.7 ps/(nm·km)) destroys the phase matching. Dispersion-shifted fibre (D ≈ 0) would make the same link unusable. The 8 × 7 = 56 interacting channel pairs also explain why DCF modules should have D ≠ 0 locally and why unequal or interleaved polarisation can add further suppression.

4.3 SBS and SRS thresholds

Using $A_{eff} = 80\ \mu\text{m}^2 = 8 \times 10^{-11}\ \text{m}^2$ and $L_{eff} = 21.17\ \text{km} = 2.117 \times 10^4\ \text{m}$:

$$P_{th,SBS} \approx 21\,A_{eff}/(g_B\,L_{eff}) = 21 \times 8\times10^{-11} / (5\times10^{-11} \times 2.117\times10^4) = 1.587 \times 10^{-3}\ \text{W} = 1.59\ \text{mW}\ (+2.0\ \text{dBm})$$

$$P_{th,SRS} \approx 16\,A_{eff}/(g_R\,L_{eff}) = 16 \times 8\times10^{-11} / (6\times10^{-14} \times 2.117\times10^4) = 1.008\ \text{W}\ (+30.0\ \text{dBm})$$

Comparison	Per channel (2 mW)	Total (16 mW)	Design (1 mW; 8 mW)
SBS threshold 1.59 mW	**Exceeded** (2 mW = 1.26 × P_{th}, +1.0 dB above)	Not additive – SBS acts on each channel separately	1 mW is 2.0 dB below threshold
SRS threshold 1.008 W	0.2 % of threshold	1.6 % of threshold (63× below)	0.8 % of threshold

SBS is a single-channel effect (Brillouin gain bandwidth only 20–100 MHz), so the relevant comparison is with the **per-channel** power: 2 mW exceeds the 1.59 mW threshold of a narrow-line carrier. SRS, whose gain bandwidth spans ≈ 13 THz, couples all channels, so the **total** power is compared: 16 mW is only 1.6 % of 1.008 W. The intensity-modulated signal contains a strong carrier line: with ER = 8 dB ($P_1/P_0 = 6.31$) the carrier holds $((\sqrt{6.31}+1)/2)^2/((6.31+1)/2) = 84$ % of the power, so modulation raises the threshold by only 0.74 dB to ≈ 1.88 mW. A 2 mW channel is still above threshold.

An estimate of SRS-induced tilt across the 5.6 nm band, using a linear Raman gain slope $g_R(\Delta\nu) \approx g_R\,\Delta\nu/13.2\ \text{THz} = 3.2\times10^{-15}$ m/W at 0.7 THz, gives a power transfer $g\,P_{tot}\,L_{eff}/A_{eff} = 3.2\times10^{-15} \times 0.016 \times 2.117\times10^4 / 8\times10^{-11} \approx 1.35$ %, i.e. 0.059 dB per span and ≈ 0.35 dB over six spans – small and equalised by the amplifiers' gain-flattening filters.

4.4 SBS versus SRS

Property	Stimulated Brillouin scattering	Stimulated Raman scattering
Physical origin	Interaction with acoustic phonons (electrostrictive density wave)	Interaction with optical phonons (molecular vibrations of silica)
Direction	Backward only in SMF (reflected, ≈ 11 GHz downshift)	Forward and backward; ≈ 13 THz (≈ 100 nm) downshift
Gain bandwidth	20–100 MHz – a single channel	≈ 7 THz wide – all channels
Peak gain	$g_B = 5\times10^{-11}$ m/W (large)	$g_R = 6\times10^{-14}$ m/W (small)
Threshold (this link)	1.59 mW per channel	1.008 W total
Effect on DWDM	Limits per-channel power; reflected power adds noise, saturates and distorts signal	Power transfers from shorter to longer wavelengths (tilt), crosstalk
Role of linewidth	P_{th} rises as $(1 + \Delta\nu_s/\Delta\nu_B)$; a MHz DFB (< 20 MHz) gains nothing; dither or modulation broadens	Insensitive to the laser linewidth

4.5 Eye safety and mitigation methods

Total power of 8 × 2 mW = 16 mW = 10 $\log_{10}(16)$ = **12.04 dBm**. Against the assumed 10 mW (+10 dBm) lowest-class limit this is 1.6 times (2.04 dB) too high: the 2 mW design would need Class 1M/3R-type controls and restricted access. Reducing to 1 mW per channel gives 8 mW = 9.03 dBm, 0.97 dB below the limit, and also brings every channel 2.0 dB below the SBS threshold. Per channel, 1.25 mW would still just meet the limit (10 mW total), but the margin would be only 1.04 dB for SBS.

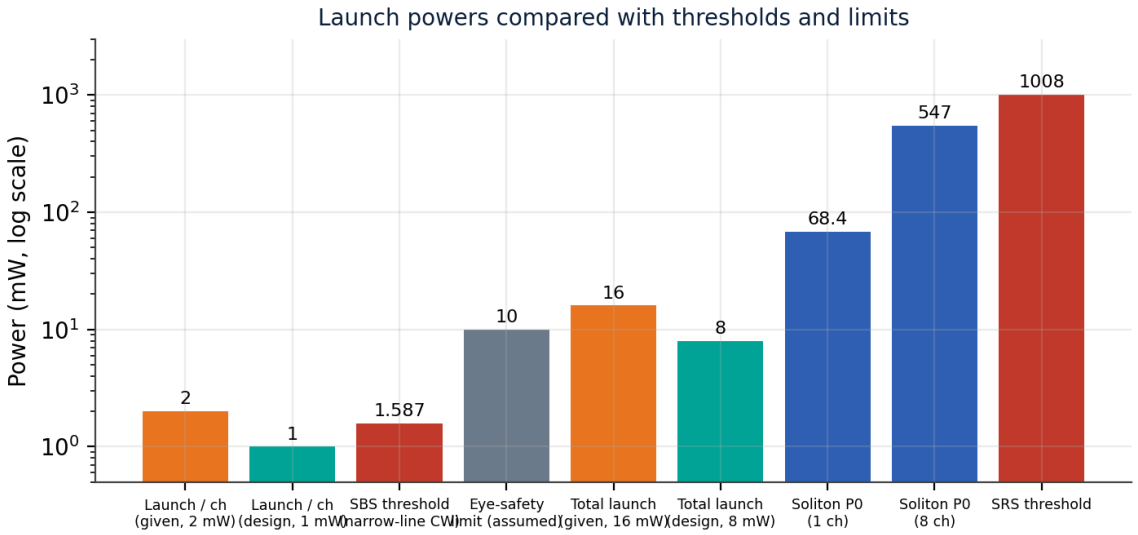


Figure 6 – Launch powers against the SBS, SRS and eye-safety limits (log scale).

The price of lowering the launch power from 2 mW to 1 mW is an OSNR reduction of 3 dB. The link OSNR (0.1 nm reference bandwidth) after N amplified spans is OSNR = 58 + P_{ch} − L_{span} − NF − 10 log N:

Launch / ch	P (mW)	OSNR (dB)	φ_{NL}, 6 spans (rad)	SBS margin (dB)	Total power (dBm)
−3 dBm	0.5	25.2	0.083	+5.0	6.0
0 dBm (design)	1.0	28.2	0.165	+2.0	9.0
+1 dBm	1.25	29.2	0.206	+1.0	10.0
+3 dBm (given)	2.0	31.2	0.330	−1.0	12.0

For example at 0 dBm: OSNR = 58 + 0 − 16 − 6 − 10 log 6 = 58 − 22 − 7.78 = 28.2 dB. This is about 8 dB above the assumed 20 dB requirement, so the 3 dB given up is not significant, whereas going the other way (2 mW) would breach the SBS and eye-safety limits.

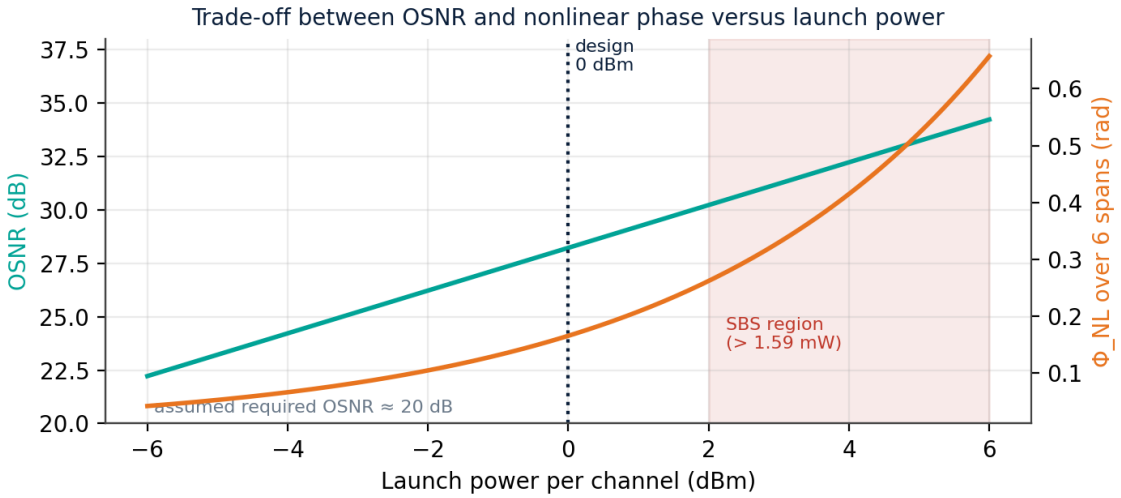


Figure 7 – OSNR and six-span nonlinear phase versus launch power; the design point is 0 dBm per channel.

Two practical methods for reducing nonlinear penalty without unnecessarily lowering the received signal:

- **SBS suppression by laser dither / phase modulation.** A low-index sinusoidal dither of the bias current (or a phase modulator) at tens to hundreds of MHz broadens the optical line beyond $\Delta\nu_B$, raising the SBS threshold by 5–10 dB so that the per-channel power can be raised toward 2 mW if needed, without lowering the received level.
- **Dispersion-managed span with local dispersion kept high plus an optimised launch.** Leaving the fibre at D = 15.7 ps/(nm·km) between amplifiers, and compensating with DCF only inside the amplifier mid-stage at low power, keeps phase matching low ($\eta_{FWM} \approx 3.4\times10^{-5}$) and walk-off high ($L_w$ = 3.2 km), while the launch is set at the OSNR/NL optimum. Additional measures are polarisation interleaving of adjacent channels and unequal channel spacing.

DESIGN DECISION 3 – Nonlinearity-control strategy

I recommend **a per-channel launch of 0 dBm (1 mW, 8 mW total), local-dispersion-managed spans with in-amplifier DCF, SBS dither as spare margin, and automatic laser shutdown** nonlinear-control strategy because the SPM phase is only 0.165 rad over six spans, XPM and FWM are strongly reduced by 12.6 ps/km walk-off and 7.9 rad/km phase mismatch (≈ −39 dB FWM crosstalk), the SBS threshold of 1.59 mW (1.88 mW with the modulated carrier) is exceeded by the given 2 mW but not by 1 mW (2.0 dB margin), SRS (1.008 W) is irrelevant, and the 8 mW total complies with the assumed 10 mW eye-safety limit while the OSNR of 28.2 dB remains well above the requirement.

Stage 5 – Soliton Transmission, Digital Backpropagation and Photonic Integration

5.1 Physical balance behind soliton transmission

In the anomalous-dispersion regime ($\beta_2 < 0$, which holds for standard fibre at 1550 nm) the dispersive chirp is of the opposite sign to the chirp produced by Kerr self-phase modulation. For a sech-shaped pulse of the correct peak power the two effects cancel exactly and the pulse propagates without change of shape. The governing nonlinear Schrödinger equation is

$$\partial A/\partial z + (i\beta_2/2)\,\partial^2 A/\partial T^2 = i\gamma|A|^2A - (\alpha/2)A$$

with the soliton order $N^2 = L_D/L_{NL}$, $L_D = T_0^2/|\beta_2|$ and $L_{NL} = 1/(\gamma P_0)$. A fundamental soliton requires $N = 1$, i.e. $L_D = L_{NL}$, which fixes the peak power $P_0 = |\beta_2|/(\gamma T_0^2)$. If $N \ll 1$, dispersion dominates and the pulse simply spreads; if $N > 1$ higher-order solitons breathe periodically.

5.2 Soliton calculations

Dispersion length

$$L_D = T_0^2/|\beta_2| = (15\text{ ps})^2 / 20\text{ ps}^2/\text{km} = 225/20 = 11.25\text{ km}$$

Fundamental-soliton peak power (L_{NL} = L_{D})

$$P_0 = 1/(\gamma\, L_D) = 1/(1.3\text{ W}^{-1}\text{km}^{-1} \times 11.25\text{ km}) = 0.0684\text{ W} = 68.4\text{ mW}\ \ (+18.35\text{ dBm})$$

Nonlinear length and soliton order at 2 mW

$$L_{NL} = 1/(\gamma P_0) = 1/(1.3 \times 0.002) = 384.6\text{ km}$$

$$N^2 = L_D/L_{NL} = 11.25/384.6 = 0.02925 \;\rightarrow\; N = 0.171$$

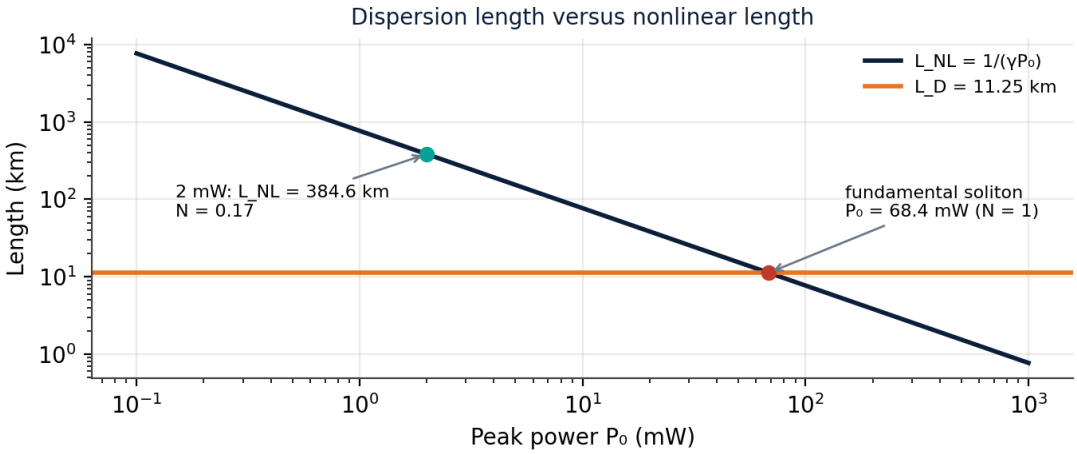


Figure 8 – Dispersion length versus nonlinear length; N = 1 requires 68.4 mW.

Because $N = 0.171 \ll 1$, the 2 mW pulses are in the **dispersion-dominated linear regime**. At 1 mW N is even smaller (0.121). A pulse of $T_0 = 15$ ps would broaden over one 80 km span ($z/L_D = 7.11$) by a factor $\sqrt{(1 + 7.11^2)} = 7.18$, to ≈ 108 ps, which is 2.7 bit periods.

Soliton condition	Required	Available / calculated	Satisfied?
Soliton order N	≈ 1	0.171 (2 mW), 0.121 (1 mW)	No
Peak power per channel	68.4 mW (+18.4 dBm)	2 mW given; 1 mW design	No (34 × short)
Total power, 8 channels	547 mW (+27.4 dBm)	16 mW	No; > eye safety, > SBS

Soliton condition	Required	Available / calculated	Satisfied?
Amplifier spacing vs L_D (path averaging)	$L_{amp} \ll L_D$ ($\ll$ 11.25 km)	80 km = 7.1 L_D	No
Soliton period $z_0 = (\pi/2)L_D$	$\gg L_{amp}$ preferable	17.7 km $\ll$ 80 km	No
Bit slot / T_{FWHM} (interaction)	$\gtrsim 5$ (T_{FWHM} = 1.763 × 15 = 26.4 ps)	40 ps / 26.4 ps = 1.5	No – strong interaction

Conclusion: soliton transmission is **not feasible** for this link. The launch power would have to be 34 times larger per channel; eight channels would require 547 mW (+27.4 dBm), beyond the eye-safety limit, far above the SBS threshold of 1.59 mW, and with a 80 km amplifier spacing that violates the path-averaged-soliton condition. Soliton collisions between WDM channels would add timing jitter. The link is therefore operated in the linear, dispersion-managed NRZ regime.

5.3 Digital backpropagation (DBP)

DBP is a digital emulation of the link run in reverse. The received complex field A(L) is propagated numerically backwards through the inverse nonlinear Schrödinger equation, usually by the split-step Fourier method, alternately applying the inverse linear operator (dispersion $e^{+i\beta_2\omega^2\Delta z/2}$, loss) and the inverse nonlinear phase rotation $e^{-i\gamma|A|^2\Delta z_\{eff}\}$. If the model matches the link, chromatic dispersion and the deterministic part of the Kerr nonlinearity are removed together.

Practical limitations. (i) DBP needs the complex field, i.e. a coherent receiver with ADCs, and the proposed link uses intensity modulation with direct detection, in which phase is lost. (ii) It compensates only the channels that are digitally captured; XPM and FWM from the other seven channels are unknown to a single-channel receiver, so single-channel DBP gives only a modest reach gain (typically 0.5–1.5 dB), and compensating all eight channels needs an ADC capturing the full 700 GHz band and heavy computation (many steps per span × 6 spans). (iii) Signal–ASE interaction, PMD and polarisation effects are not reversible.

Appropriateness: DBP is **not appropriate for the complete 8-channel link** because the receiver is non-coherent, the nonlinear phase is already small (0.165 rad over six spans at the design power) and the computational cost is out of proportion. Per-span DCF plus optimal launch removes the need. If the system were upgraded to coherent DP-QPSK/16QAM, electronic dispersion compensation (not DBP) would normally be sufficient, and DBP would be reserved for a high-power, single-super-channel upgrade.

5.4 Standards parameters and compliance check

Parameter	ITU-T G.957 / G.959.1 / G.694.1	IEEE 802.3bm / bs (per 25G lane)	Proposed transmitter
Central wavelength / grid	Range per application code; G.694.1 100 GHz grid, 193.1–193.8 THz	Lane wavelength window (LAN-WDM 800 GHz or CWDM, as per PHY)	1546.9–1552.5 nm on 100 GHz grid; ±0.1 nm
Mean launched power	Max/min per code; per-channel limits in G.698.x	Average launch power and OMA per lane	0 dBm per channel; 8 mW total
Spectral width	RMS or −20 dB width max	SMSR and spectral window	Single mode, SMSR ≥ 35 dB
SMSR	≥ 30 dB	≥ 30 dB	> 35 dB (DFB)
Extinction ratio	Minimum (8.2 dB for STM-16 class; 8–10 dB DWDM)	Minimum ER (≈ 4–5 dB depending on PHY); TDECQ	≥ 8 dB (EAM)
Eye mask / TDECQ	Eye mask, rise/fall	Transmitter eye mask, jitter, TDECQ	Within mask; 17.5 GHz BW
RIN, return loss	Max reflectance, RIN	$RIN_{12}OMA \leq -130$ dB/Hz (class dependent); return-loss tolerance	Isolator, RIN $\lesssim -140$ dB/Hz
Safety	Laser safety, ALS (G.664)	Class 1 by IEC 60825-1	8 mW = +9.03 dBm; ALS
FEC	Optional (G.975.1)	RS(528,514) or RS(544,514)	RS-FEC in host

Note: G.957 is written for SDH links up to STM-16 (2.5 Gb/s); it is used here for its parameter framework (launch power, spectral width, SMSR, ER, eye mask), with G.694.1 for the grid and G.959.1/G.698.x for DWDM, while IEEE 802.3bm/bs provide the 25 Gb/s-per-lane electrical and optical requirements. Exact numerical limits must be taken from the standard edition adopted by the project.

How the transmitter is checked: wavelength, SMSR and spectral width with an optical spectrum analyser at 0, 25 and 40 °C; average and OMA power with a calibrated power meter; extinction ratio, eye mask and TDECQ with a sampling oscilloscope and a reference receiver using a PRBS31 pattern; RIN with a calibrated RIN set-up; return-loss tolerance with a controlled reflection; ALS by interrupting the fibre; and total power against the hazard limit at the multiplexer output. Burn-in and temperature cycling verify long-term wavelength stability.

5.5 Discrete versus InP versus Si-photonics

Criterion	Discrete (8 TOSAs + external mux)	Monolithic InP (8 DFB-EAM + combiner + SOA)	Si-photonics + hybrid III–V laser
Footprint	Largest (8 packages, 8 TECs)	Smallest active footprint: one chip, one TEC	Small; photonics compact but needs bonded lasers
Power consumption	Highest (8 TECs, 8 drivers)	Low – one TEC, EAM low drive (≈ 2 Vpp)	Lowest for modulators at scale; MZM drive and laser coupling losses cost power
Coupling losses	8 laser-to-fibre + mux joints	One output fibre; on-chip waveguide	Fibre and laser-to-Si coupling loss 1–3 dB each
Reliability	Many fibre joints and solder points	Fewer joints; mature InP lasers; one bad laser = chip failure	Hybrid bonding maturity still growing
Scalability	Poor – linear in channel count	Good to 10–40 λ; yield falls with channel count	Excellent in volume; CMOS foundry
Wavelength tracking	Independent drifts	AWG and lasers on one thermal stage track together	Si has low thermo-optic coefficient (0.07 nm/°C) – laser still drifts
Cost	Lowest NRE, highest unit cost	High NRE, medium unit cost for moderate volume	Highest NRE, lowest unit cost at high volume

Integration influences system performance in four ways. First, removing separate coupling and mux joints saves roughly 1.5–3 dB per channel, which directly increases the OSNR budget or lowers the pump power. Second, lasers and multiplexer on one chip drift together (≈ 0.1 nm/°C), so the single TEC holds both and eases the ±0.1 nm requirement. Third, fewer discrete parts raises MTBF. Fourth, thermal and electrical cross-talk between neighbours (heat of ≈ 40 mW per laser, RF coupling) must be managed by layout and ground design, and is the main risk of integration.

DESIGN DECISION 4 – Transmitter implementation

I recommend **a monolithic InP 8-channel DFB-EAM transmitter PIC (single TEC, on-chip combiner and SOA/booster stage, discrete EDFA booster, in-span DCF; no soliton, no digital backpropagation, RS-FEC only)** transmitter/integration strategy because DFB sources at 193.1–193.8 THz are native to InP, the EAM provides > 30 GHz bandwidth and low chirp, the single TEC holds the lasers and combiner within ±0.1 °C, the integrated approach saves 1.5–3 dB of coupling loss per channel and reduces footprint and part count, soliton operation is infeasible ($N = 0.171$, $P_0 = 68.4$ mW), and DBP is unnecessary at 0.165 rad of nonlinear phase and impossible with direct detection. Si-photonics with hybrid lasers becomes preferable at high volume or for 400G+ co-packaged modules.

Stage 6 – Verification and Critical Evaluation

The results of Stages 2–5 are compared with the original requirement below.

Requirement	Specified / Target	Achieved	Satisfactory?
Optical source selection	Single-mode, ≈ 1550 nm, narrow linewidth, high speed	DFB with EAM; λ = 1.55 µm (193.4 THz); η_{int} = 0.80; SMSR > 35 dB	Yes
Modulation bandwidth and output power	≥ 17.5 GHz; adequate power for 6 spans	20 GHz (+2.5 GHz, 14.3 %), EAM > 30 GHz; P = 8.4 mW (9.24 dBm); 0 dBm/ch after losses	Yes (margin from EAM)
Spectral / chirp and wavelength stability	±0.1 nm over 40 °C; low chirp	Free-running drift 4.0 nm (5 slots) reduced to ±0.01 nm (TEC ±0.1 °C) + locker; total ±0.07 nm; chirp 0.02 nm (EAM) vs 0.109 nm (DML)	Yes
Nonlinear-impairment control and eye safety	SBS < threshold, $\varphi_{NL} \ll 1$ rad, total ≤ 10 mW	SBS 1.59 mW vs 1 mW (−2.0 dB margin); φ_{NL} = 0.165 rad; SRS 1.008 W vs 8 mW; XPM/FWM ≈ −39 dB; 8 mW = 9.03 dBm	Yes
Soliton / dispersion condition	N ≈ 1 for soliton operation; else compensate dispersion	L_D = 11.25 km; P_0 = 68.4 mW; N = 0.171 → soliton not feasible; DCF compensates 1256 ps/nm per span	Soliton: No; linear DM link: Yes
Integration / DSP / standards strategy	Compact, reliable; G.957 / IEEE 802.3bm-bs compliant	Monolithic InP PIC, one TEC; RS-FEC only; DBP rejected; parameters mapped to standards and test plan	Yes

6.1 Consistency of the results with the requirement

The calculations support one another. The wavelength and efficiency results (1.55 µm, 0.80) justify the DFB; its 8.4 mW and 20 GHz bandwidth cover the power and speed budget; the 4 nm drift drives the TEC and locker; and the EAM resolves the chirp and bandwidth margin. The launch power fixed in Stage 4 (1 mW per channel) is simultaneously consistent with the SBS threshold (1.59 mW), the eye-safety limit (8 mW vs 10 mW), a six-span SPM phase of 0.165 rad, and an OSNR of 28.2 dB, so no requirement has been satisfied by sacrificing another. The soliton and DBP analyses show that the link works in the linear regime, which is exactly what the NRZ direct-detection design assumes.

6.2 Additional checks

- **PMD:** with 0.1 ps/√km, DGD ≈ 0.1 × √480 = 2.2 ps, below the ≈ 0.1 T_b = 4 ps tolerance of 25 Gb/s NRZ.
- **Spectral occupancy:** the signal main lobe (±25 GHz) uses half of each 100 GHz slot, giving a spectral efficiency of 0.25 b/s/Hz and ≈ 50 GHz guard between lobes.
- **Dispersion map:** 1256 ps/nm per span is cancelled by ≈ 12.6 km DCF (−100 ps/(nm·km)); residual dispersion per channel (slope mismatch) must be < ±100 ps/nm.

6.3 Critical evaluation – limitations and risks

- **Assumptions:** required OSNR (20 dB), NF (6 dB), α_H, Brillouin bandwidth and the 10 mW eye-safety limit are assumed; their values should be re-measured on the final hardware.
- **SBS margin (2.0 dB)** is modest for a carrier-dominated NRZ spectrum; the dither option is therefore provided as spare margin.
- **DCF** adds loss (≈ 6 dB per module), nonlinearity (smaller effective area) and ripple; a chirped fibre Bragg grating could replace it.
- **Integration risk:** a single failed laser or EAM in the monolithic chip forces replacement of the entire PIC; spare channels or a pluggable fallback reduce the risk.

- **Calculations not performed:** amplifier gain tilt and ripple cascading, filter narrowing from the AWG, polarisation-dependent loss and reflection-induced RIN have only been treated qualitatively.

Final Transmitter Design – Integrated Engineering Solution

7.1 Final design / parameter table

Parameter	Adopted value	Basis
Channels / grid	8, 193.1–193.8 THz (1552.52–1546.92 nm), 100 GHz	ITU-T G.694.1
Source	1550 nm DFB (λ = 1.24/0.80 = 1.55 µm), integrated EAM (EML)	Stage 2
Internal quantum efficiency	0.80 (τ_r = 10 ns, τ_{nr} = 40 ns)	Stage 2
Laser bias / threshold	40 mA / 12 mA (I/I_{th} = 3.33)	Given
Laser output power	8.4 mW (9.24 dBm), η_s = 0.3 mW/mA	Stage 3
EAM bias / swing	≈ −1.5 V / 2 V peak-to-peak (assumed); ER ≥ 8 dB	Stage 3
Modulation bandwidth	20 GHz laser, > 30 GHz EAM; required 17.5 GHz	Stage 3
Data format	25 Gb/s NRZ, PRBS31; RS-FEC in host	IEEE 802.3bm/bs
TEC set-point and stability	25 °C, ±0.1 °C (±0.01 nm); locker ±0.02 nm	Stage 3
Launch power / channel	0 dBm (1 mW); total 8 mW (9.03 dBm)	Stage 4
SBS / SRS / eye-safety margins	2.0 dB / 21 dB / 0.97 dB	Stage 4
Nonlinear phase	0.0275 rad/span; 0.165 rad over 6 spans	Stage 4
Dispersion map	1256 ps/nm per span cancelled by ≈ 12.6 km DCF	Stage 3/5
Soliton parameters	L_D = 11.25 km; P_0 = 68.4 mW; N = 0.171 (not feasible)	Stage 5
OSNR at receiver	28.2 dB (0.1 nm); requirement ≈ 20 dB	Stage 4
Integration	Monolithic InP PIC with a single TEC; EDFA booster	Stage 5

7.2 Block diagram

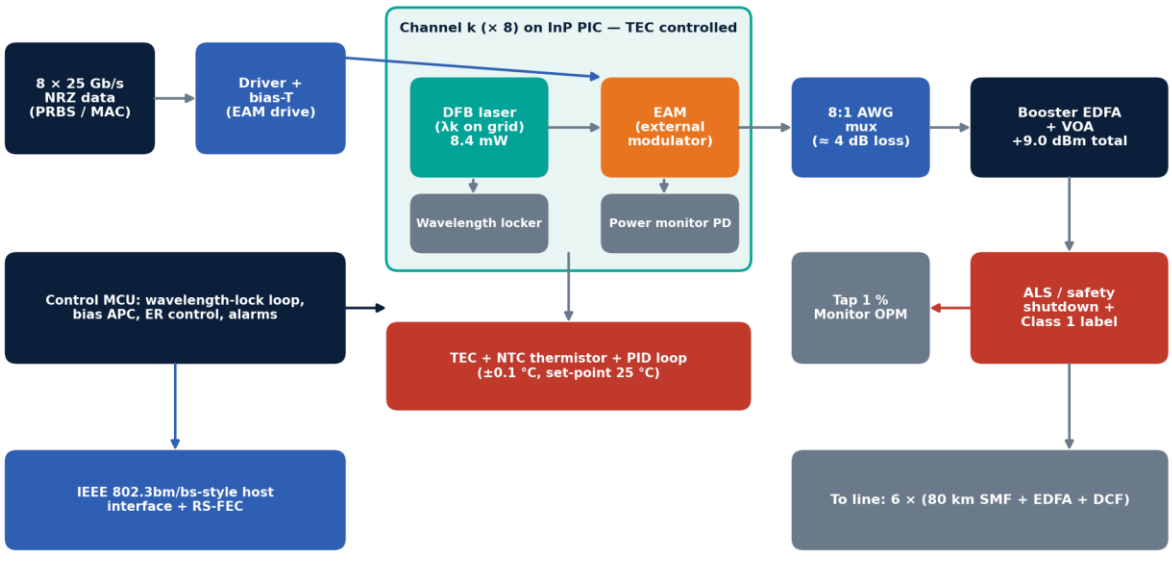


Figure: block diagram of the proposed 8-channel DWDM transmitter

Figure 9 – Proposed 8-channel DWDM transmitter: DFB–EAM array on a TEC-controlled InP PIC, AWG multiplexer, booster EDFA and safety shutdown.

7.3 How the choices work together

Modulation and wavelength control. Each channel is a DFB laser with a monolithically integrated EAM. The 40 mA DFB bias provides 8.4 mW and 20 GHz bandwidth, the EAM supplies > 30 GHz and low chirp. The TEC holds the chip at 25 °C within ±0.1 °C; a wavelength locker (periodic etalon) measures the residual error and trims the TEC set-point. The 4.0 nm free-running drift is thus reduced to ±0.01 nm, well within ±0.1 nm and ≪ the 0.8 nm spacing.

Nonlinear, eye-safety and compensation strategy. The multiplexer output is boosted so that each channel is launched at 0 dBm. This keeps SBS 2.0 dB below threshold, SPM at 0.165 rad over six spans, SRS ≈ 21 dB below threshold and the aggregate power (9.03 dBm) below the 10 dBm limit, with an automatic laser shutdown. XPM and FWM are suppressed by the 12.6 ps/km walk-off and 7.9 rad/km phase mismatch of the SMF, and the 1256 ps/nm dispersion per span is removed by in-amplifier DCF.

Integration and standards. The monolithic InP implementation removes eight laser-to-fibre joints and tracks all lasers and the combiner on one TEC. Its parameters (SMSR, ER, OMA, eye mask, wavelength, RIN, power) are mapped to ITU-T G.957/G.959.1/G.694.1 and IEEE 802.3bm/bs and verified with the test plan of Section 5.4.

Soliton feasibility. With L_D = 11.25 km and N = 0.171 at 2 mW (P_0 would need to be 68.4 mW, 547 mW for eight channels), soliton transmission is infeasible. The link therefore uses dispersion-managed linear NRZ transmission.

7.4 Final engineering recommendation

The recommended transmitter is a monolithic InP 8-channel DFB-EAM array with TEC and wavelength-locker control, launching 0 dBm per channel into a dispersion-managed SMF link with EDFAs. Starting from the requirement it follows that the source must emit at 1.55 µm with η_{int} = 0.80, which a DFB does; the modulation must exceed 17.5 GHz with low chirp and the temperature drift of 4 nm must be reduced to ±0.01 nm, which an EAM and TEC achieve; the launch power must keep SBS (1.59 mW), SPM (0.165 rad) and the eye-safety limit (10 mW) under control, which 1 mW per channel does; and the soliton and DBP options are rejected by calculation. Every subsystem is consistent with the others, and the OSNR of 28.2 dB leaves about 8 dB margin.

FINAL ENGINEERING DECISION – Would I deploy the transmitter solution I created?

YES. The transmitter meets every quantitative requirement derived in this report: wavelength 1.55 µm (193.4 THz, η_{int} = 0.80), power 8.4 mW (9.24 dBm), bandwidth margin 2.5 GHz (EAM > 30 GHz), wavelength stability ±0.07 nm worst case against ±0.1 nm, φ_{NL} = 0.165 rad, SBS margin 2.0 dB, SRS margin 21 dB, total power 9.03 dBm < 10 dBm and OSNR 28.2 dB > 20 dB, and it maps cleanly to the ITU-T and IEEE parameter sets. Deployment is conditional on (1) confirming the Class 1 limit and FEC threshold against the current standard editions, (2) verifying the 20 dB OSNR requirement and DCF slope matching on a pilot link, (3) acceptance testing of SMSR, ER, TDECQ and wavelength error across 0–40 °C, and (4) fitting automatic laser shutdown. Soliton transmission and digital backpropagation are deliberately not deployed.

Appendix A – Summary of Equations and Numerical Results

All formulas used in the report are collected here with the substituted values, so that each result can be checked independently.

No.	Equation	Substitution	Result
A1	$\lambda = 1.24/E_g$	1.24/0.80	1.55 μm; 193.4 THz
A2	$\eta_{int} = \tau_{nr}/(\tau_r+\tau_{nr})$	40/(10+40)	0.80
A3	$\eta_d = \eta_s \cdot q/h\nu$	0.3 W/A ÷ 0.80 V	37.5 %
A4	$P = \eta_s(I - I_{th})$	0.3 × (40 − 12)	8.4 mW = 9.24 dBm
A5	$B = 0.7\ R_b$	0.7 × 25	17.5 GHz (margin 2.5 GHz)
A6	$f_{3dB}(I) = 20\sqrt{[(I - 12)/28]}$	≥ 17.5 GHz	I ≥ 33.4 mA
A7	$\Delta\nu = (\alpha_H/4\pi)(1/P)dP/dt$	(3/4π)/17.5 ps	13.6 GHz = 0.109 nm
A8	$\Delta T = D\ L\ \Delta\lambda$	15.7 × 80 × 0.109	137 ps
A9	$L_{max} = 104\ 000/(D\ B^2)$	104 000/(15.7 × 625)	10.6 km
A10	$\Delta\lambda = (d\lambda/dT)\Delta T$	0.1 × 40	4.0 nm = 5 slots
A11	$\Delta T_{TEC} = \text{tolerance}/(d\lambda/dT)$	0.1/0.1	±1 °C (design ±0.1 °C)
A12	$L_{eff} = (1 - e^{-\alpha L})/\alpha$	(1 − 0.0251)/0.04605	21.17 km
A13	$\varphi_{NL} = \gamma\ P\ L_{eff}$	1.3 × 0.002 × 21.17	0.055 rad; 0.330 rad (6 spans)
A14	$L_w = T_b/(D\ \Delta\lambda)$	40/(15.7 × 0.8)	3.2 km
A15	$\Delta\beta = 2\pi\lambda^2 D(\Delta f)^2/c$	(100 GHz spacing)	7.9 rad/km
A16	$\eta_{FWM} = \alpha^2/(\alpha^2 + \Delta\beta^2)$	$0.0461^2/(0.0461^2 + 7.9^2)$	3.4×10^{-5}
A17	$P_{th,SBS} = 21A_{eff}/(g_B L_{eff})$	$21 \times 8\times10^{-11}/(5\times10^{-11} \times 2.117\times10^4)$	1.59 mW
A18	$P_{th,SRS} = 16A_{eff}/(g_R L_{eff})$	$16 \times 8\times10^{-11}/(6\times10^{-14} \times 2.117\times10^4)$	1.008 W
A19	$OSNR = 58 + P - L - NF - 10 \log N$	58 + 0 − 16 − 6 − 7.78	28.2 dB
A20	$L_D = T_0^2/\lvert\beta_2\rvert$	225/20	11.25 km
A21	$P_0 = 1/(\gamma L_D)$	1/(1.3 × 11.25)	68.4 mW
A22	$L_{NL} = 1/(\gamma P)$	1/(1.3 × 0.002)	384.6 km
A23	$N = \sqrt{(L_D/L_{NL})}$	√(11.25/384.6)	0.171

Appendix B – Notation and Worked OSNR Derivation

Symbol	Meaning	Symbol	Meaning
E_g	Bandgap energy	α_H	Linewidth-enhancement factor
τ_r, τ_{nr}	Radiative / non-radiative lifetime	f_R	Relaxation-oscillation frequency
η_{int}, η_d	Internal / differential efficiency	D, β_2	Dispersion parameter / GVD
I_{th}, η_s	Threshold current / slope efficiency	γ, A_{eff}	Nonlinear coefficient / effective area
L_{eff}	Effective fibre length	g_B, g_R	Brillouin / Raman gain coefficients
φ_{NL}	Nonlinear phase shift	L_D, L_{NL}	Dispersion / nonlinear length
N	Soliton order (or channel count)	T_0, P_0	Pulse width / soliton peak power
SMSR	Side-mode suppression ratio	OSNR	Optical signal-to-noise ratio

The OSNR relation used in Stages 4 and 6 follows from the amplified-spontaneous-emission noise of a chain of N identical spans. Each EDFA with gain $G = L_{span}$ adds ASE power in the 0.1 nm reference bandwidth (12.5 GHz) of $P_{ASE} = NF \cdot h\nu \cdot B_{ref} \cdot G$, where NF is the noise figure, $B_{ref} = 12.5$ GHz and $h\nu = 1.28 \times 10^{-19}$ J at 1550 nm. Thus

$$h\nu \cdot B_{ref} = 1.28\times10^{-19}\ \text{J} \times 1.25\times10^{10}\ \text{Hz} = 1.6 \times 10^{-9}\ \text{W} = -58\ \text{dBm}$$

The signal-to-ASE ratio after N amplifiers is therefore (in dB)

$$\text{OSNR} = P_{ch} - (-58\ \text{dBm}) - L_{span} - NF - 10\log_{10} N = 58 + P_{ch} - L_{span} - NF - 10\log_{10} N$$

For $P_{ch} = 0$ dBm, $L_{span} = 16$ dB, NF = 6 dB and N = 6, $10\log_{10} 6 = 7.78$ dB and the OSNR is $58 + 0 - 16 - 6 - 7.78 = 28.2$ dB. The figure of 58 dB is the quantum limit with 0 dB loss and 0 dB noise figure, from the photon energy at 1550 nm and the 0.1 nm bandwidth.

References and standards consulted

- ITU-T Recommendation G.957, Optical interfaces for equipments and systems relating to the synchronous digital hierarchy; G.959.1, Optical transport network physical layer interfaces; G.694.1, Spectral grids for WDM applications: DWDM frequency grid; G.698.x, Multichannel DWDM applications with single-channel optical interfaces; G.664, Optical safety procedures and requirements.
- IEEE Std 802.3bm (40 Gb/s and 100 Gb/s fibre optic Ethernet) and IEEE Std 802.3bs (200 Gb/s and 400 Gb/s Ethernet).
- ANSI Z136.2, Safe Use of Optical Fiber Communication Systems Utilizing Laser Diode and LED Sources; IEC 60825-1 and 60825-2.
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